Cloudmon Installation Guide

This document provides step-by-step instructions for partners to install and configure **Cloudmon** on Ubuntu systems.

Contents

1. Cloudmon Controller Installation (Node.js & NGINX Included)

This document provides step-by-step instructions for partners to install and configure **Cloudmon** on Ubuntu systems, including the **Controller**, **MongoDB**, **Probe**, and **nProbe (NTM)** components.

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1. Overview

1.1 Install Node.js 18.12 LTS

```
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt-get install -y nodejs
```

1.2 Install NGINX (v1.20+)

```
sudo apt update
sudo apt install -y curl gnupg2 ca-certificates lsb-release

echo "deb http://nginx.org/packages/mainline/ubuntu $(lsb_release -cs) nginx" | sudo tee /etc/apt/sources.list.d/nginx.list
curl -fsSL https://nginx.org/keys/nginx_signing.key | sudo apt-key add -
sudo apt update
sudo apt install -y nginx
```

1.3 Install Cloudmon Controller

1. Copy the **Controller package URL** from:

   https://repo.veryxtech.com/cloudmon/versions.json

2. Download and install:

```
curl -O <controller-package-url>
chmod +x cloudmon-controller-<version>-Linux-x86_64.run
./cloudmon-controller-<version>-Linux-x86_64.run
```

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2. MongoDB Installation and Configuration

2.1 Install MongoDB Community Edition

Follow the official MongoDB documentation:

https://www.mongodb.com/docs/v7.0/tutorial/install-mongodb-on-ubuntu/

2.2 Create Cloudmon Database and User

```
mongosh
use cloudmon

db.createUser({
  user: "cloudmon",
  pwd: "<password>",
  roles: [ { role: "readWrite", db: "cloudmon" } ]
})
```

> **Note:** Ensure the password contains **no special characters**.

2.3 Enable MongoDB Authorization

Edit the MongoDB configuration:

```
sudo vi /etc/mongod.conf
```

Add under `security`:

```
security:
  authorization: enabled
```

Restart MongoDB:

```
sudo service mongod restart
```

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3. Configure Cloudmon Database and Package URL

Edit the Cloudmon environment file:

```
sudo vi /usr/local/cloudmon/server/.env
```

Update the following values:

```
DATABASE_URL="mongodb://cloudmon:<password>@<server-ip>:27017"
DATABASE="cloudmon"
PACKAGE_URL="https://repo.veryxtech.com/cloudmon/versions.json"
```

Restart Cloudmon Controller:

```
sudo service cloudmon-controller@* restart
```

Verify status:

```
sudo service cloudmon-controller@* status
```

MongoDB Logrotate Configuration (Optional)

> **Note:** Perform this step **only when MongoDB is installed on a separate VM** (not on the same VM as the Cloudmon Controller).

4.1 Create Logrotate File

```
sudo vi /etc/logrotate.d/mongodb
```

Add the following:

```
/var/log/mongodb/mongod.log {
    daily
    size 10M
    rotate 10
    dateext
    compress
    delaycompress
    missingok
    notifempty
    postrotate
        if pidof mongod >/dev/null; then
            kill -SIGUSR1 $(pidof mongod)
        fi
    endscript
}
```

4.2 Test Logrotate Configuration

```
sudo logrotate -d /etc/logrotate.d/mongodb
```

4.3 Verification

```
ls -lh /var/log/mongodb/
```

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5. Access Cloudmon UI

Open a browser and access:

```
https://<controller-ip>
```

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6. Cloudmon MIB Browser Installation (Controller)

- **Document:** Cloudmon_MIB_Browser-Readme
- **Download:** http://repo.veryxtech.com/cloudmon/downloads/snmp-mib-browser.zip

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3. Cloudmon Probe Installation

This section describes installation and configuration of the **Cloudmon Probe** on a separate VM or host.

3.1 Install Node.js 18.12 LTS

```
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt-get install -y nodejs
node -v
```

3.2 Install Dependencies

```
sudo apt-get install -y libpcap-dev nmap
nmap
```

3.3 Install Cloudmon Collector & Probe

Download and install the **Collector** and **Probe** packages from the Cloudmon UI:

Help → Probe Page

3.4 Node.js Version Issue (Node 12)

If Node.js 12 is installed, refer to:

https://github.com/nodesource/distributions/issues/1157

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4. nProbe (NTM) Installation and Configuration

4.1 Install nProbe

Download from:

Index of /apt-stable

> **Pre-requisite:** Ensure outbound port **8883** is open for cloud licensing.

4.2 Collect nProbe System Information

```
nprobe -v
```

Capture the **Version**, **Build OS**, and **SystemID**.

4.3 Apply nProbe License

```
echo "<license-string>" > /etc/nprobe.license
```

Send license request to: **presales@cloudmon.ai**

4.4 Verify Traffic Reception

```
sudo tcpdump -i <interface> udp <port>
```

(Default NetFlow port: **2055**)

4.5 Create nProbe Configuration File

```
sudo nano /etc/nprobe/nprobe-port2055.conf
```

Add:

```
-3="2055"
--verbose=0
-P="/var/log/nprobe-2055/"
-T=%EXPORTER_IPV4_ADDRESS %INPUT_SNMP %OUTPUT_SNMP %IPV4_SRC_ADDR %IPV4_SRC_MASK %IPV4_DST_ADDR %IPV4_DST_MASK %L4_SRC_PORT %L4_SRC_PORT_MAP %L4_DST_PORT %L4_DST_PORT_MAP %SRC_TOS  %APPLICATION_NAME %L7_PROTO_NAME %L7_PROTO_CATEGORY %PROTOCOL %PROTOCOL_MAP %IN_BYTES %IN_PKTS %FIRST_SWITCHED %LAST_SWITCHED %DIRECTION %SAMPLING_INTERVAL %NPROBE_IPV4_ADDRESS %SRC_AS_MAP %DST_AS_MAP %POST_NAT_SRC_IPV4_ADDR %POST_NAT_DST_IPV4_ADDR %POST_NAPT_SRC_TRANSPORT_PORT %POST_NAPT_DST_TRANSPORT_PORT %FLOW_PROTO_PORT %FLOW_START_MILLISECONDS %FLOW_END_MILLISECONDS
```

4.6 Create Log Directory and Permissions

```
sudo mkdir /var/log/nprobe-2055
sudo groupadd nprobe
sudo usermod -aG nprobe nprobe
sudo chown nprobe:nprobe /var/log/nprobe-2055/
```

4.7 Update Cloudmon Collector Environment

Edit:

```
sudo vi /usr/local/cloudmon/collector/.env
```

Add:

```
FLOW_PATH='/var/log/nprobe-2055'
```

Restart Collector Service.

4.8 Start and Verify nProbe Service

```
sudo service nprobe@port2055 status
sudo service nprobe@port2055 restart
```

4.9 nProbe Debugging

- Verify firewall allows **UDP 2055**
- Ensure NetFlow traffic is received
- Check log files under `/var/log/nprobe-2055/`
